02 - Problem Statement and Objectives

From the project's own documentation

Knowledge-pack rule: every statement below is derived from files in this project or is explicitly labelled as planned, missing or not verified. Nothing here is an invented citation, dataset, result or feature.

Source files used for this document

documentation/project_overview.md; documentation/research_questions.md; report/chapters/chapter_1_introduction.md

Problem statement

- Fraud-detection research often optimises predictive metrics on a single dataset and reports accuracy, which is misleading when fraud is ~2% of transactions.
- Consumer-trust research relies on surveys that are disconnected from measured security performance.
- Detection errors have a consumer cost: a false positive blocks or challenges a legitimate customer.
- Research pipelines are rarely observable: results are not monitored or easily reproduced.

Aim

Build and evaluate a reproducible platform that links fraud and threat detection with consumer trust and monitors the experiment.

Objectives

ID	Objective	Where addressed in the code
O1	Reproducible pipeline for transaction and security-event data	src/pipeline.py, src/data_collection, src/preprocessing
O2	Compare supervised and unsupervised detectors with imbalance-aware metrics	src/fraud_detection/train.py, src/anomaly_detection/detectors.py, src/model_evaluation/evaluate.py
O3	Quantify the cost of detection errors	false_positive_rate metric, risk tiers (src/fraud_detection/risk.py), notebook 07
O4	Measure consumer trust with a privacy-conscious survey	survey/questionnaire.md, src/trust_scoring/scoring.py, src/preprocessing/anonymize.py
O5	Propose and stress-test a composite Consumer Trust Index	scoring.consumer_trust_index, notebook 05 sensitivity analysis
O6	Make the pipeline observable in Grafana with real data	src/metrics/exporter.py, dashboard/

Research questions and hypotheses

ID	Research question
RQ1	Which supervised model gives the best fraud trade-off (PR-AUC, recall, false-positive rate)?
RQ2	How far can label-free detectors (Isolation Forest, LOF) get compared with supervised learning?
RQ3	What do detection errors cost the consumer at different thresholds?
RQ4	Which signals characterise each fraud pattern and which patterns evade detection?

ID	Research question
RQ5	How do consumers perceive FinTech cybersecurity and which factors relate to trust?
RQ6	Can perceived trust and measured security be combined into a stable index?
RQ7	Can pipeline health and model quality be monitored with open-source observability tools?

Hypotheses H1-H7 (amount differs for fraud; channel and authentication associated with fraud; trust differs by age; past fraud experience lowers trust; boosted trees beat Logistic Regression on PR-AUC; MFA users report higher perceived security) are listed in documentation/research_questions.md.

Status: hypotheses are tested only on synthetic data in this repository. No hypothesis has been tested on real consumers.

Scope and ethics

- Security experiments use synthetic data inside an isolated Docker network, or an optional licensed public dataset the user downloads.
- Survey design: voluntary, no free text, banded demographics, salted-hash pseudonyms, k-anonymity (k=5).
- Ethics approval is listed as a required checkbox in survey/questionnaire.md and is NOT recorded as obtained.